
Decomposing LLM-Judge Uncertainty to Target Expert Labels

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Abstract

An LLM judge evaluates outputs at scale. Experts should label only where it is least sure. Its natural escalation signal conflates two uncertainties: *aleatoric*, real disagreement in the expert pool, which labels cannot reduce, and *epistemic*, the judge’s ignorance, which labels do reduce. A small Bayesian model separates them: a regression on labels already collected learns how far to trust a black-box judge’s prediction. Both components follow as simple formulas, with no sampling or further judge calls. The components isolate on a real LLM judge against exactly known truth, and stated confidence is no guide to its actual error. On real human disagreement (ChaosNLI) the epistemic ranking removes 83% more error than total uncertainty for the same expert labels, though simply escalating the least-labelled items does as well there. We demonstrate we can estimate where a judge is ignorant rather than where experts genuinely disagree, and propose using this to direct expert labelling.

1 Introduction

LLMs increasingly serve as judges: evaluators substituting for expert labels at scale. Judges are stochastic. Averaging repeated calls is the standard remedy [Lail and Markham, 2026], but treats all variation as noise, when only part of it is. The *pool* of human experts the judge stands in for may read an item differently, so the judge should predict a *distribution* over labels, not a single verdict [Baan et al., 2022; Leonardelli et al., 2025]. Its spread is *aleatoric* uncertainty, a property of the item no further labelling removes. The judge may also be ignorant of where the pool’s distribution sits: *epistemic* uncertainty.

This paper asks whether the two can be disentangled for a black-box judge, from its outputs and the labels already collected.

Contributions. (1) *A model of judge uncertainty separating the two components*: a small regression on already-collected labels learns how far to trust the judge’s prediction on each item. Both components follow as simple formulas with no further judge calls. The entropy of the predicted distribution, the natural escalation score, is exactly the two components summed, which is why ranking on it conflates ignorance with disagreement. (2) *A controlled test against exact ground truth*: a frontier LLM judges dice whose true bias and pool disagreement are known, with the judge black-box. The components isolate, and the judge’s stated confidence, taken at face value, is no guide to its actual error. (3) *A real-world check on human disagreement*: on ChaosNLI the epistemic ranking recovers more label-budget value than total uncertainty, though not more than a count-based rule.

2 Related work

Calibration to majority labels is ill-posed under genuine disagreement [Baan et al., 2022; Plank, 2022]. LeWiDi-2025 [Leonardelli et al., 2025] standardises evaluation against the pool’s distribution. Distributional judging [Chen et al., 2025; Wang et al., 2025] supplies the prediction our model consumes, without per-item uncertainty about it. Systems that escalate on undivided judge-side uncertainty target a single verdict [Jung et al., 2025; Sheng et al., 2025; Badshah et al., 2026]. Bachar et al. [2026] carry the distinction only as self-reported flags, and Dorner et al. [2025] prove an aggregate limit on judge value. Our framing and the classical decomposition are the same object, not rivals [Kotelevskii et al., 2025], and our epistemic score is BALD’s [Houlsby et al., 2011; Bickford Smith et al., 2023; Nguyen et al., 2019]. Dubois et al. [2026] fit generalised linear models to grader scores: theirs separates grader severity offsets from residual disagreement, while ours separates irreducible pool disagreement from our own ignorance of where the pool sits. The mechanism is theirs. The transfer, failure characterisation, and judge-side measurement are ours. None asks which labels to buy.

3 Model

The model builds up in three levels: a single annotator (level 0), the pool of annotators (level 1), and our knowledge of the pool (level 2).

Levels 0 and 1: an annotator, and the pool. Annotator a in a pool with weights w_a holds a label distribution $\pi_a(x)$ over the K classes, and a label is a draw from it. A label from a randomly drawn annotator is then a draw from the pool’s *population label distribution* $p_x = \sum_a w_a \pi_a(x)$. Its entropy $H(p_x)$ (or Gini $G(p_x) = 1 - \|p_x\|^2$) is the *aleatoric* component: a property of the pool, unreduced by labels. We model the pool at this marginal level (real-valued scores: App. B).

Level 2: our knowledge of the pool. p_x is never observed, so we hold a belief about it. An estimate’s realised error is a divergence $D_x = d(\hat{p}_x, p_x)$. *Epistemic* uncertainty is a prediction of D_x made without observing p_x . d is the Manhattan distance $\sum_j |\hat{p}_j - p_j|$, running 0 (exact) to 2 (probability on disjoint classes), standard for categorical soft labels [Leonardelli et al., 2025]. Correctly predicting a genuinely split pool gives $D_x = 0$: disagreement is not error.

The estimator. The level-2 belief is a Dirichlet centred on an LLM judge’s predicted distribution $\hat{q}_x$, weighted by a *trust* λ_x in pseudo-label units. The n_x labels in hand, tallied c_x , update it in closed form to $\text{Dir}(\alpha_x)$ with mean $m_x = \alpha_x / \alpha_{0,x}$ and *total evidence* $\alpha_{0,x} = \lambda_x + n_x$, trust plus labels. Trust is fitted log-linearly on item features, among them embedding distance/density to labelled items, the *familiarity channel*, a signal not from the judge. Ordinary labels are the only training signal (full displays and likelihood: App. B). The judge’s prediction is never fitted: the judge does the perceptual work, our model decides how far to believe it.

The decomposition. A fresh label’s uncertainty splits as

$$H(m_x) = \mathbb{E}[H(p_x) \mid c_x] + I(y; p_x \mid c_x),$$

with y one further label. The aleatoric term is closed-form (App. B). The epistemic term I_x is BALD’s acquisition score [Houlsby et al., 2011], and total uncertainty $H(m_x)$ is the escalation baseline. Entropy sees only the shape of m_x , not the evidence behind it: with equal total evidence the two rankings nearly agree, and where evidence varies they come apart, up to full inversion (App. B).

4 Experimental setup

Two experiments share one real LLM judge, Azure OpenAI’s gpt-5.6-terra, queried 18 August 2026.¹

¹Code, data, both raw judge caches, and the pre-registration records: <https://github.com/composo-ai/judge-uncertainty-decomposition>.

Table 1: Component isolation: Spearman rank correlation of each estimate with the true quantities, 500 dice. The truths are themselves anti-correlated (-0.23), so the separation is not inherited from the world. Full grid: App. C.

estimate	vs. disagreement	vs. error
aleatoric	0.45	-0.12
epistemic (I)	0.08	0.66

Table 2: Escalation on ChaosNLI at budget $B = 10\%$ (the top tenth of items escalated), Manhattan units of error removed. All seven rules: App. D.

rule	value	vs. entropy
epistemic (I -ranking)	15.0	$+6.8 [+1.7, +11.9]$
fewest-labels-first	14.3	$+6.1 [-0.0, +12.4]$
entropy (total uncertainty)	8.2	—

Experiment 1: dice (exactly known truth). 500 three-sided dice with biases p_i carry descriptions in four tiers (none, weak, directional, strong), assigned independently of bias. From the description alone the judge predicts the 100-roll distribution (three prompt variants, 0–100 stated confidence), with 0–10 revealed rolls as labels. With p_i known, true disagreement is its entropy, true error the posterior’s Manhattan distance from it. The regression is given each die’s tier as its familiarity feature. We measure whether each estimated component tracks its ground truth, and whether stated confidence does.

Experiment 2: ChaosNLI (real human disagreement). 3,113 items [Nie et al., 2020], each a premise–hypothesis sentence pair labelled entailment, neutral, or contradiction by 100 annotators. The judge predicts the annotator distribution. Labels are half-split per item, fitting and acquisition on one half, evaluation the other: evaluation truth is never touched by training, and disjoint 50-label halves set a 0.158 noise ceiling (Manhattan). We measure the estimation error each escalation rule removes for a fixed escalation budget B , the fraction of items sent for expert labels: every rule escalates its top-ranked $B\%$, so rules compete purely on which items they pick.

Escalation rules. *Ours* ranks by the estimated epistemic component: mutual information I_x under log loss and, under squared error, the one-label value Δ_x , with posterior spread v_x the withhold variant (App. B). Experiment 2 reports all three. Baselines: *entropy* (output or posterior), *verbalised confidence*, *fewest-labels-first*, *random*, *oracle*.

5 Results

5.1 Simulations: the estimator on synthetic pools with known truth

Two simulations precede the judge experiments: one calibrated, one misspecified (setup and the misspecified results: App. A). The calibrated one draws worlds from the Sec. 3 model itself, so the value of every escalation choice is exactly computable. When items differ only in how many labels they already hold, ranking by total uncertainty forfeits a third of the error reduction the same labels could have bought, while epistemic ranking stays at oracle level. When evidence is equal the gap is exactly zero, as the Sec. 3 claim requires. Posterior spread alone captures nearly all of that gain, so the takeaway is simple: escalate on the epistemic component (numbers: App. A, Figure 2).

5.2 Experiment 1: a real LLM judge on dice with exactly known truth

On a real LLM judge the two components are separable: each estimated component tracks its own ground truth and not the other’s (Table 1). Figure 1 shows it in motion.

Stated confidence fails as an escalation signal: near-certainty with no information, near-zero when merely told the dice are weighted, calibrated only near complete information (Table 3). At item level it correlates *positively* with error (Spearman $+0.22$), and escalating on low confidence does no better than random (App. C).

Dice are the wrong world for measuring escalation gains, which is why the decision-level claim waits for Experiment 2: this judge predicts uniform exactly when ignorant, so ranking by total uncertainty nearly coincides with ranking by ignorance, and between-rule differences drown in single-draw noise (App. C). ChaosNLI supplies the regime where the two come apart.

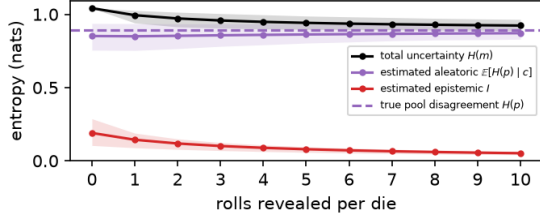


Figure 1: The decomposition in motion (500 dice, 200 draws, 2.5–97.5 bands). The aleatoric estimate holds at 0.85–0.87 against the true 0.892 (dashed), the epistemic falls 0.190 to 0.052, and total uncertainty falls to the aleatoric floor: labels drain ignorance, not disagreement.

Table 3: The confidence pathology by tier: stated confidence (0–100) is non-monotone in judge error. Full four-column table: App. C.

tier	judge error	stated confidence
none	0.53	90.9
weak	0.54	0.4
directional	0.41	59.1
strong	0.007	14.8

5.3 Experiment 2: escalation on real human disagreement (ChaosNLI)

On ChaosNLI the judge is confidently wrong again: its predicted annotator distributions miss at triple the noise ceiling, the irreducible error from annotator sampling (Sec. 4), while its stated confidence averages above 90 on the 0–100 scale. Neither confidence nor prediction entropy tracks error (App. D).

The regression finds near-uniform trust, each prediction worth about five labels, unsurprising for short crowdsourced NLI pairs, which give the familiarity features little to exploit. But label counts range 0–10 by design, leaving total evidence α_0 heterogeneous where λ is flat (the Sec. 3 condition).

On real human disagreement the epistemic ranking recovers more label-budget value than total uncertainty, though a count-based rule does as well here. For the same label budget, epistemic ranking removes 15.0 units of error against entropy’s 8.2 (Table 2): 83% more per expert label. The gap widens at larger budgets and is absent at smaller ones. The rules split into count-aware and count-blind clusters, and the count heuristic is not beaten (App. D).

The Sec. 3 mechanism, measured in Sec. 5.1, is present on real data. Separating our estimator from the count heuristic needs items genuinely differing in judge familiarity, with dense labels to score. No public dataset offers both: the field’s missing dataset rather than this paper’s missing experiment.

6 Limitations and conclusion

We demonstrate the full method only on synthetic dice, where the truth is exactly known. On ChaosNLI the judge was about equally unfamiliar with every item, so its fitted trust (λ) came out nearly flat, and only the label-count side of the model was really tested on real data. That side worked, removing nearly twice as much error as entropy ranking at the same budget (Sec. 5.3). But when trust is flat our ranking follows label counts, exactly what the fewest-labels-first baseline does, so the baseline ties us. The remaining limits are more ordinary. Each experiment used one judge model and one prompt design. Simulated effect sizes depend on the generator (Sec. 5.1). Our intervals resample labels over a fixed item set and say nothing about new items. We have no defence against a deceptive judge that games the familiarity signals behind its trust. A merely worthless signal reduces the method to ranking by label counts, still ~ 32 pp better than entropy ranking in the simulations (Sec. 5.1).

Escalate the items the judge is ignorant about, not the items annotators genuinely disagree on. The two can be separated with the labels already collected. No new data are needed. Escalating on ignorance removed more error per expert label than escalating on total uncertainty, though not more than a fewest-labels-first rule. Overall, this demonstrates a first step towards decomposing epistemic and aleatoric uncertainty in LLM judges. The decisive test is on data where judge familiarity genuinely varies, with the comparison against fewest-labels-first stated before any results are seen.

References

- Baan et al. Stop measuring calibration when humans disagree. In *EMNLP*, 2022.
- Bachar et al. LLM performance predictors: Learning when to escalate in hybrid human–AI moderation systems. In *AAMAS*, 2026.
- Badshah, Emami, and Sajjad. SCOPE: Selective conformal optimized pairwise LLM judging. In *ICML*, 2026. arXiv:2602.13110.
- Bickford Smith et al. Prediction-oriented Bayesian active learning (EPIG). In *AISTATS*, 2023.
- Chen et al. Distributional LLM-as-a-judge. In *NeurIPS*, 2025.
- Donner, Nastl, and Hardt. Limits to scalable evaluation at the frontier: LLM as judge won’t beat twice the data. In *ICLR*, 2025.
- Dubois, Coppock, et al. Quantifying biases in LLM-as-a-judge evaluations. In *ICML*, 2026.
- Houlsby et al. Bayesian active learning for classification and preference learning. *arXiv:1112.5745*, 2011.
- Jung, Brahman, and Choi. Trust or escalate: LLM judges with provable guarantees for human agreement. In *ICLR*, 2025.
- Kotelevskii et al. From risk to uncertainty: Generating predictive uncertainty measures via Bayesian estimation. In *ICLR*, 2025.
- Lail and Markham. On cost-effective LLM-as-a-judge improvement techniques. In *ICML 2026 Workshop on Statistical Frameworks for Uncertainty in Agentic Systems*, 2026. arXiv:2604.13717.
- Leonardelli et al. LeWiDi-2025 at NLPerspectives: Third edition of the learning with disagreements shared task. In *NLPerspectives @ EMNLP*, 2025. arXiv:2510.08460.
- Nguyen, Destercke, and Hüllermeier. Epistemic uncertainty sampling. In *Discovery Science*, 2019.
- Nie, Zhou, and Bansal. What can we learn from collective human opinions on natural language inference data? In *EMNLP*, 2020. ChaosNLI.
- Plank. The “problem” of human label variation: On ground truth in data, modeling and evaluation. In *EMNLP*, 2022.
- Sheng et al. Analyzing uncertainty of LLM-as-a-judge: Interval evaluations with conformal prediction. In *EMNLP*, 2025. arXiv:2509.18658.
- Wang, Zhang, and Choi. Improving LLM-as-a-judge inference with the judgment distribution. In *Findings of EMNLP*, 2025. arXiv:2503.03064.

Appendices

A Supporting simulations in full

Terms used for escalation results here and in the body: a rule’s *regret* at budget B is its top- $B\%$ selection’s shortfall below the best possible, in percentage points (pp). *Advantage* is the difference of two rules’ regrets. Escalation *value* is the reduction in error from purchased labels, in Manhattan units, summed over the escalated set. Real-judge numbers are means over 200 label re-draws with 95% intervals, paired within re-draw.

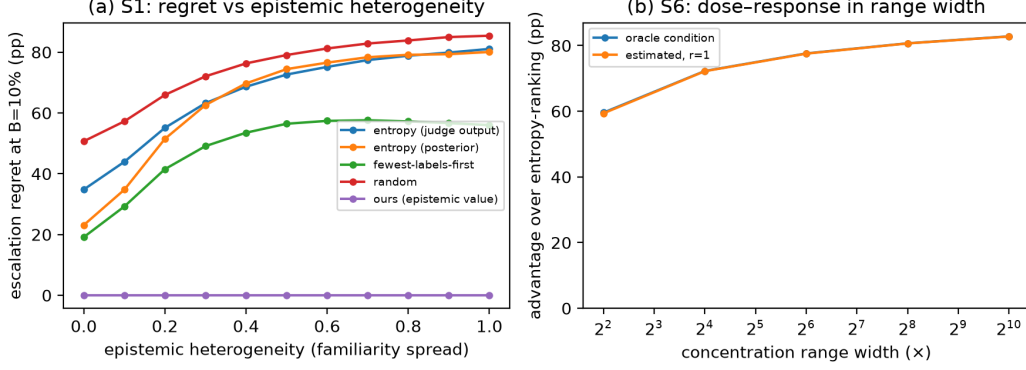


Figure 2: (a) Escalation regret at $B = 10\%$ as the spread of item trust grows. Entropy-ranking (on the judge’s output, or on the posterior with labels folded in) degrades from 35pp toward random’s 85pp, fewest-labels-first captures part of the gap and plateaus, and ranking by estimated label value stays at oracle level. (b) The effect does not require an extreme world: as the trust range narrows from $\times 256$ to $\times 4$, the advantage over entropy-ranking declines only from 81pp to 60pp, and the estimated-trust condition matches the oracle condition at every width.

Setup. In the *calibrated* simulation (2,000 items), the world is drawn from the Sec. 3 model with the judge’s prior correct by construction, so the value of every escalation decision is exactly computable. The spread of item trust, the spread of pool disagreement, and the correlation between them are steerable axes, and each item reveals 0–10 labels. In the *misspecified* simulation, the world is drawn truth-first, the judge’s prediction quality varies with familiarity, and its stated confidence is corrupted by a knob γ : honest at $\gamma = 0$, information-free at $\gamma = 1$, inverted at $\gamma = 1.5$. There the estimator’s model is wrong by construction and rules are scored against an oracle that knows the realised world, so only between-rule comparisons are meaningful.

Calibrated simulation: all four findings. Pass/fail criteria: advantage ≥ 15 pp at the criterion cell and oracle tracking Spearman ≥ 0.9 , measured 80.9pp (seed-level CI 80.6–81.2) and 1.0. (1) The failure needs no exotic regime: with identical prior trust and only revealed counts (0–10) differing, entropy-ranking loses 34.8pp. In the true null (identical trust *and* identical label count), the advantage measures 0.0pp, as the Sec. 3 claim requires. (2) Entropy-ranking approaches random as trust spread grows (81pp regret vs random’s 85pp at the widest setting), because entropy is blind to evidence and, through revealed counts, anti-correlated with label value: the claim’s inversion, realised. (3) There is no estimator-quality threshold, and adverse correlation does not close the gap: with a worthless familiarity signal the regression still beats entropy-ranking by 32pp (revealed counts alone carry the signal), rising monotonically to 81pp as the signal improves, and when contentious items are also the unfamiliar ones (correlation 0.8, the realistic direction), the advantage is still 50pp. (4) Ranking by posterior spread captures $\sim 98\%$ of the gain (v -ranking within 1.2–1.8pp of Δ -ranking everywhere). The finer withhold-versus-purchase distinction (App. B) is second-order for ranking.

Misspecified simulation: the predictions as scored. The prediction that a regression using only revealed-label counts survives all tested corruptions was refuted at $\gamma = 1$ and confirmed elsewhere: the collapse requires both uninformative confidence and silently varying quality. The prediction that the regression would monotonically down-weight stated confidence as corruption grows was refuted in its monotone form: the fitted coefficient grows through $\gamma = 0.75$, reaches zero at $\gamma = 1$, and flips sign at inversion. The regression re-learns the mapping rather than gradually distrusting it (Figure 3). These magnitudes are generator-dependent: they do not predict real-world effect sizes.

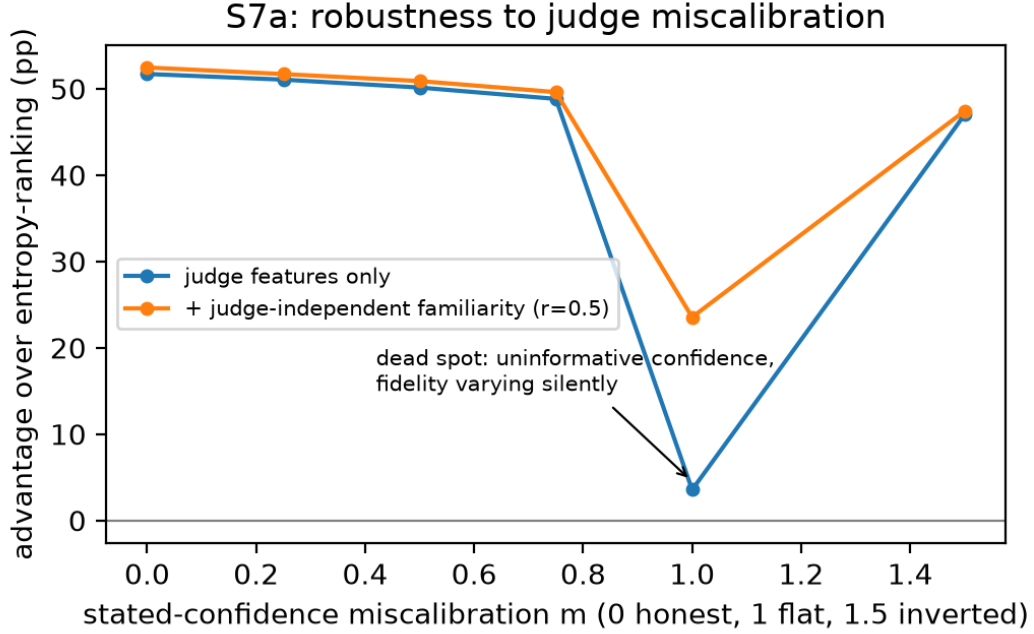


Figure 3: Advantage over entropy-ranking at $B = 10\%$ in the misspecified simulation, as stated-confidence corruption γ varies (the axis label reads m , this paper’s earlier name for the knob). With judge-supplied features only (blue), the advantage is flat at $\sim 50\text{pp}$ from honest through badly miscalibrated ($\gamma \leq 0.75$) and at fully inverted confidence ($\gamma = 1.5$), where the regression re-learns the mapping and exploits the inversion (fitted coefficient on stated confidence flips from -0.63 to $+1.09$). The one dead spot is $\gamma = 1$: confidence carrying no information while prediction quality varies silently (3.6pp). A judge-independent familiarity signal of middling quality (orange) recovers most of it (23.5pp) and costs nothing elsewhere.

B Model details: the estimator, fitting likelihood, closed forms, and the real-valued formulation

The estimator in full. An LLM judge supplies a prediction $\hat{q}_x$ of the pool. The belief, the labels in hand (n_x of them, class tally c_x), and the trust, fitted log-linearly on item features, are

$$p_x \sim \text{Dir}(\lambda_x \hat{q}_x), \quad c_x \sim \text{Multinomial}(n_x, p_x), \quad \log \lambda_x = \beta_0 + \beta^\top z_x,$$

with z_x collecting variant spread, stated confidence, the label count n_x itself, and embedding distance/density to labelled items. The coefficients β are the only fitted quantities, and the log keeps λ_x positive. The posterior is

$$p_x \mid c_x \sim \text{Dir}(\alpha_x), \quad \alpha_x = \lambda_x \hat{q}_x + c_x, \quad \alpha_{0,x} = \lambda_x + n_x,$$

labels and trust in one currency: updating is addition.

The real-valued formulation. Annotator a holds a latent score $\mu_a(x) \in [0, 1]$, the score they would settle on given unlimited care, and an observed label adds within-annotator noise, which we take to be Gaussian,

$$y = \mu_a(x) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma_a^2),$$

where σ_a^2 is the variance of a fresh rating of the same item by the same annotator. A pool of annotators $a = 1, \dots, M$ with weights w_a induces a population score distribution with mean and spread

$$m(x) = \sum_a w_a \mu_a(x), \quad \tau^2(x) = \sum_a w_a (\mu_a(x) - m(x))^2,$$

where $\tau^2(x)$ is genuine inter-annotator disagreement, a property of the pool itself, unreduced by collecting further labels.

Neither $\mu_a(x)$ nor $\tau^2(x)$ is directly observable: a label carries the annotator’s own noise with it. Averaging labels still recovers $m(x)$, since that noise is centred, but a label’s variance is $\tau^2(x) + \bar{\sigma}^2$ with $\bar{\sigma}^2 = \sum_a w_a \sigma_a^2$, so one label per (item, annotator) pair cannot separate genuine disagreement from within-annotator noise. It need not: both are aleatoric, and every quantity in the paper uses only their sum.

For real-valued scores, the split displayed in Sec. 3 is the three-way variance decomposition

$$\text{Var}(y \mid x, \mathcal{D}) = \underbrace{\bar{\sigma}^2}_{\text{within-annotator}} + \underbrace{\mathbb{E}[\tau^2 \mid \mathcal{D}]}_{\text{between-annotator}} + \underbrace{\text{Var}(m \mid \mathcal{D})}_{\text{epistemic}},$$

whose first two terms are aleatoric (irreducible for a single fresh label) and whose third alone shrinks with labels. The entropy display of Sec. 3 is its categorical counterpart, with the first two terms merged by the Level-0 identifiability constraint above.

The fitting likelihood. The maximum-likelihood objective for β is the Dirichlet–multinomial likelihood

$$\mathcal{L}(\beta) = \prod_{x: n_x \geq 1} \text{DirMult}(c_x \mid n_x, \lambda_x(\beta) \hat{q}_x),$$

a generalised linear model with log link on λ and Dirichlet–multinomial response (beta–binomial regression generalised past two classes). A handful of coefficients against a few hundred labelled items is well-posed: only the trust scale is learned.

Closed form of the aleatoric term. $\mathbb{E}[H(p_x) \mid c_x] = \psi(\alpha_{0,x} + 1) - \sum_j m_{x,j} \psi(\alpha_{x,j} + 1)$, with ψ the digamma function.

The derived scores. Under squared error the decomposition yields two decision quantities: the posterior spread $v_x = G(m_x)/(\alpha_{0,x} + 1)$, the expected squared error of the current estimate (the *withhold* quantity), and the one-label value

$$\Delta_x = \frac{G(m_x)}{(\alpha_{0,x} + 1)^2},$$

the expected reduction in that error from one further expert label (the *purchase* quantity). Withholding and purchasing optimise different scores [Bickford Smith et al., 2023], both distinct from entropy. In this model class the two rank items nearly identically (Sec. 5.1), and the claim the paper tests is the two-way split, epistemic against total.

When entropy-ranking is safe. Entropy sees only the shape of m_x , but label value also sees the evidence behind it, decaying as $(\alpha_{0,x} + 1)^{-2}$. With equal total evidence the two rankings nearly agree. Where it varies they come apart, folded-in labels driving entropy anti-correlated with value, up to full inversion. Equal evidence is the only α_0 condition giving agreement for every shape, and it fails in any partially-labelled deployment: label counts alone make α_0 heterogeneous.

C Experiment 1 auxiliary results

All queries in both experiments used the Azure OpenAI deployment `gpt-5.6-terra`, API version 2024-12-01-preview.

Escalation numbers on dice. Total predictive entropy tracks judge error at 0.57, against 0.66 for the epistemic estimate. Escalating on low stated confidence is indistinguishable from random ($B = 10\%$: +1.8pp, per-draw 95% interval -14 to $+18$). Ours-versus-entropy regret differences are dominated by single-draw noise (+6pp expected, per-draw -25 to $+28$).

Self-updating versus conjugate updating. When shown its own rolls, the judge’s self-updated prediction (error 0.258, $n = 380$) loses to conjugate updating of its own prior (adding the observed rolls to its stated prediction as counts) once the familiarity channel sets the trust level (0.262 with the channel, 0.307 without): the Bayesian machinery is where the correction lives, not overhead.

Table 4: Isolation of the estimated components. Spearman correlation between each estimated component and each ground truth, with 95% intervals over label re-draws (real judge) or simulated worlds (calibrated). Isolation requires a strong diagonal and near-zero off-diagonal. The truth–truth column shows the two ground truths are themselves uncorrelated or anti-correlated, so the separation is not inherited from the world.

world	est-aleatoric vs		est-epistemic vs		truth–truth
	true disagr.	true error	true disagr.	true error	
calibrated simulation	0.77 (.76,.78)	0.13	−0.14	0.48 (.46,.51)	−0.03
real LLM judge	0.45 (.32,.55)	−0.12	0.08	0.66 (.51,.70)	−0.23

Table 5: The judge’s stated confidence is non-monotone in its error. By description tier (125 dice each): judge error is the mean Manhattan distance from the judge’s predicted face distribution to the true one (0–2 scale). Stated confidence is the judge’s self-reported certainty on its 0–100 scale, and variant spread is the standard deviation of the predicted face probabilities across three prompt paraphrases, in probability units (0.0002 means the paraphrases agree to the fourth decimal place).

tier	judge error	stated confidence	variant spread
none	0.53	90.9	0.0002
weak	0.54	0.4	0.0008
directional	0.41	59.1	0.0018
strong	0.007	14.8	0.0006

D Experiment 2 auxiliary results

Descriptives. The judge’s mean Manhattan error is 0.458 against the 0.158 ceiling, and stated confidence averages 90.3. Item-level Spearman with error is -0.13 for stated confidence and 0.11 for prediction entropy. The fitted trust is near-uniform, $\text{sd}(\log \hat{\lambda}) = 0.22$ against the planned condition’s ≥ 0.5 , each prediction worth about five pseudo-labels. The planned condition tested only the trust side and so predicted no separation, a mis-specification: the characterisation is about α_0 , not about λ alone.

Comparisons beyond Table 2. The planned headline, ranking by Δ , was inconclusive against entropy ($+4.5$ $[-2.2, +9.4]$), and the counts-only claim was refuted ($+6.9$, grazing zero at -0.4). Seven rules are compared, so no interval stands alone. I -ranking beats entropy by $+12.8$ $[+4.3, +20.4]$ at $B = 20\%$, with smaller budgets inconclusive. The value clusters are count-aware 12.1–15.0 and count-blind 6.9–8.3, and I minus fewest-labels-first is $+0.7$ $[-6.3, +7.1]$.

Table 6: Escalation on ChaosNLI at $B = 10\%$ (311 items). All entries are in Manhattan units of error removed, not percentages. Value is the total reduction from the purchased labels, summed over the escalated set, and advantage is the difference from entropy-ranking on the judge’s output, paired within replicate. Means over 200 label re-draws, Monte-Carlo standard error in parentheses, per-draw 95% intervals in brackets. Only the Δ row was the planned headline comparison. The remaining advantages are exploratory.

escalation rule	value (SE)	advantage over entropy [per-draw 95%]
ours: mutual information I	15.0 (0.2)	+6.8 $[+1.7, +11.9]$
fewest-labels-first	14.3 (0.2)	$+6.1$ $[-0.0, +12.4]$
ours: label value Δ (<i>planned headline</i>)	12.7 (0.2)	$+4.5$ $[-2.2, +9.4]$
ours: posterior spread v	12.1 (0.2)	$+3.9$ $[-1.7, +10.7]$
random	8.3 (0.2)	$+0.1$ $[-5.5, +5.6]$
entropy (judge output)	8.2 (0.1)	—
verbalised confidence	8.1 (0.1)	-0.1 $[-5.8, +5.1]$
entropy (posterior)	6.9 (0.1)	-1.2 $[-6.0, +3.3]$